



Module 19 Day 2: Advanced Security Monitoring Tools

19.2 Lesson Plan: Advanced Security Monitoring Tools

Overview

In today's class, you will learn about advanced monitoring tools that security professionals use to monitor and analyze security events. Then we will explore career paths and certifications related to Splunk and SIEM skills. We will conclude by introducing Splunk certifications.

Class Objectives

By the end of class, you will be able to:

- Differentiate between various advanced security monitoring solutions, such as SOARs, UBAs, and UEBA's, and determine which is most appropriate for a specific security situation.
- Understand how knowledge of SIEM software and Splunk is valued in the information security job market.
- Continue learning about Splunk with free Splunk training courses.

01. Introduction to Advanced Security Monitoring Tools (0:15)

Today we will explore other information security products used by security organizations.

- We will begin by briefly introducing Splunk's SIEM product, Enterprise Security, and exploring several other security products available in the marketplace.

- Then, we will focus on security careers relevant to the knowledge and tools learned in the last five classes.
- After the break, we will introduce Splunk certifications.
- We will conclude the class by introducing the final project, the bootCon presentation.

Splunk ES

Over the past five classes, we covered many of Splunk's capabilities and add-on applications.

- While these Splunk capabilities can assist in security investigations, Splunk has a SIEM product specifically designed to assist security professionals.

Splunk's SIEM product is called Splunk Enterprise Security (ES).

- Splunk ES provides security professionals with insights from machine-generated data, such as:
 - Network devices, like routers and firewalls
 - Endpoint devices, like antivirus solutions
 - Vulnerability management systems, like Nessus

Splunk ES is one of the most popular add-on products for security professionals as it has pre-built dashboards, reports, and built-in features.

Splunk ES features allow you to:

- Identify, prioritize, and investigate security events.
- Gain insights into security events.
- Monitor the status of your security environment.
- Audit your security events.
- Navigate these tasks with a pre-built, easy-to-use interface.

Note that learning the capabilities of Splunk ES is outside the scope of this class, but you can explore the following resources in your own time if you wish to learn more:

- [Splunk: ES video demo](#)
- [Splunk: ES documentation](#)

Advanced Security Monitoring Solutions

While SIEM products such as Splunk ES provide many monitoring benefits, organizations are now integrating advanced security monitoring solutions into their businesses to provide additional protection.

Note the following about advanced security monitoring solutions:

- The core goal of all security monitoring solutions is to assist with detection of information security events.
- Advanced security monitoring solutions provide additional benefits such as machine learning, artificial intelligence, automation, and response.
- The most popular advancements in the information security industry are:
 - User behavior analytics (UBA)
 - User and entity behavior analytics (UEBA)
 - Security orchestration, automation, and response (SOAR)

UBA is a security monitoring tool that uses machine learning, artificial intelligence, and data processing to detect abnormalities in user activity.

- UBA gathers information about typical user behaviors and creates baselines.
 - For example, UBA can gather information on the servers and systems that a user accesses, as well as when and how frequently they do so.
 - UBA can then create alerts for when a user's activity deviates from this typical behavior.
 - If a user usually only logs onto a server between 9am and 5pm Monday through Friday, UBA would create an alert if the user logged on at 2am on a Saturday.

UEBA is a security monitoring tool similar to UBA, except it extends its monitoring to other "entities."

- Entities can include routers, servers, and Internet of Things (IoT) devices.
- UEBA looks at normal user and entity behaviors and creates alerts when users or entities show abnormal activity.

SOAR, comparable to a SIEM, automates security processes and responds to security incidents.

- Examples of automating security processes include:
 - Creating logging
 - Assigning priorities to security incidents
- Examples of responding to security incidents include:
 - Launching security investigations
 - Mitigating threats
- Similar to a SIEM, SOAR gathers machine data from multiple entities and analyzes the data for security events.
 - SOAR uses playbooks that detail the processes and response actions for specific events.
 - Playbooks are designed and configured by each organization.

- For example, an organization can design a playbook to automate the response to a phishing incident.
 - The image illustrates how SOAR uses a playbook to automate each step of the response.
 - This illustrates how using SOAR playbooks can decrease incident response time.

XDR stands for Extended Detection and Response.

- XDRs aim to provide a comprehensive security system that detects and responds to threats from various sources, including endpoints, networks, cloud, and email.
- XDR offers advanced detection, rapid response, and intuitive automation.
- It provides improved cross-domain threat hunting and forensic investigation capabilities.
- It lacks the log management, retention, and compliance capabilities of SIEM.

In the following activity, you will research the different vendors that offer these advanced security monitoring tools and learn about the different capabilities of each product.

02. Advanced Security Monitoring Tools (0:20)

Note the following:

- OMP management has approved a larger budget to enhance security monitoring.
- You are tasked with researching several SOAR, UBA, and UEBA vendors to provide a summary to your management about the products you are recommending.

03. Advanced Security Monitoring Tools Activity (0:10)

04. Splunk Careers (0:15)

Note that we will now examine how the Splunk knowledge and tools learned over the past five days can apply to careers and certifications in the information security industry.

- Even if an employer doesn't use Splunk, many Splunk skills can be easily applied to other vendors and similar products.
- For example, if a security professional who is familiar with Splunk ES takes a position at an organization that uses LogRhythm, they will note the similar core capabilities and quickly adapt to the different software.

Splunk skills are desirable for many InfoSec positions. For example:

- SOC analysts work in a security operations department alongside security engineers. Their positions involve detecting, containing, and potentially remediating information security threats. Most SOC analysts use SIEM products, such as Splunk ES, to monitor their environment.
- Cyber threat analysts analyze an organization's networks and applications to protect organizations from cybercriminals. Cyber threat analysts often use Splunk products to make predictions about cybercriminals and their potential attacks.
- Application security engineers can use Splunk to fix web and mobile application vulnerabilities. AppSec engineers use Splunk to analyze their application logs to assist with creating and testing their remediation.
- Network security administrators use products like Splunk to monitor suspicious network traffic, such as DDoS attacks. They can use the findings from Splunk logs to mitigate and prevent against future attacks.
- Incident response managers can use Splunk to monitor the status of ongoing security investigations when an incident has occurred.

Some of these positions, such as SOC and threat analyst, may use Splunk products as part of their primary responsibilities.

- Other security positions, such as application security administrators, may only use Splunk for a small portion of their tasks.
- Either way, understanding how to use Splunk is a valuable skill for information security professionals.

Splunk is already a required skill in many roles, and the industry demand increases every year.

(Source: Brain4ce Education Solutions, n.d.)

In the next activity, you will search through several job sites to find positions where Splunk is either a required skill or a primary responsibility.

05. Splunk Careers (0:15)

- For this activity, you will search two specific job sites for Splunk-related roles and answer provided questions about the positions you find.

06. Splunk Careers Activity (0:10)

This activity illustrated the value of Splunk in the current information security job market. You were tasked with researching jobs that require Splunk skills.

Completing this activity required the following steps:

- Using two specific provided job sites to research jobs that require Splunk skills
- Answering several questions on their findings

07. Break (0:15)

08. SIEM Certifications (0:15)

Similar to other domains in cybersecurity, Splunk skills are validated through certifications. These allow InfoSec professionals to prove they have the required skills to assist an organization.

- Certifications can help cybersecurity professionals acquire a new position or promotion and can provide networking opportunities with professionals who have similar certifications.

Note the following certifications that Splunk offers:

- Splunk Core Certified User: Entry-level certification that demonstrates a user's basic ability to use the Splunk software
- Splunk Core Certified Power User: Demonstrates a user's foundational skills with Splunk's core software, in addition to more complex skills such as creating calculated fields and data models

- Splunk Core Certified Advanced Power User: Demonstrates a user's capability to design reports, complicated searches, and dashboards
- Splunk Enterprise Certified Admin: Focuses on an individual's ability to support daily administrative tasks using Splunk Enterprise software
- Splunk Enterprise Certified Architect: Focuses on a Splunk administrator's role to support advanced troubleshooting, configurations, and deployments within Splunk Enterprise
- Splunk Enterprise Security Certified Admin: Focuses on a Splunk administrator's role to support installation, advanced troubleshooting, configurations, and deployments with Splunk ES environment

If you are interested in learning more about the tracks and certifications Splunk offers, note the following link:

- [Splunk: Training and certification](#)

Like many certifications in the InfoSec field, most of the certifications are expensive. Many of them cost over \$2,000.

Each student can independently complete any remaining training to obtain the Splunk Core Certified User Certification.

- This is completely optional.
- Note that there are costs associated with several additional courses, as well as taking the exam.
- Note the following [link](#) for any students interested in the requirements for obtaining the Splunk Core Certified User Certification.

In the final activity of the day, you will:

- Set up a Splunk account.
- Register and complete several free Splunk single-subject courses aimed at completing a certification.

Before continuing, note that other SIEM solutions have certifications as well:

- CompTIA CySA+ (Cybersecurity Analyst+): CompTIA CySA+ is an intermediate-level certification for cybersecurity professionals tasked with incident detection, prevention, and response through continuous security monitoring. The certification covers topics such as threat intelligence, security information and event management (SIEM), endpoint detection and response (EDR), and extended detection and response (XDR). CompTIA offers various training options, including eLearning, hands-on labs, and instructor-led training.
- Elastic Stack: While not a certification, Elastic Stack is a set of free, open-source tools that can help you identify and visualize cybersecurity events. Gaining hands-on experience with Elastic Stack can complement your knowledge and

skills in SIEM and other cybersecurity certifications, such as CompTIA Security+, CySA+, PenTest+, and CASP+.

Other vendor-specific SIEM certifications and training programs include:

- IBM Security QRadar SIEM
- LogRhythm SIEM
- AlienVault USM (from AT&T Cybersecurity)
- SolarWinds Security Event Manager (SEM)

In addition to SIEM certifications, there are more general security certifications, such as Certified Information Systems Security Professional (CISSP) and Certified Information Security Manager (CISM), which can provide a broader understanding of information security and may include some aspects of SIEM.

09. Splunk Certification (1:00)

Note the following:

- For the remainder of the class, you will register for and begin Splunk single-subject courses.
- You must register for a Splunk account and complete the certification registration.
- Once the registration is completed, you will complete several free single-subject courses.

10. Splunk Certification (0:05)

The goal of this activity was to introduce Splunk's free single-subject courses. You were tasked with completing several free courses.

Completing this activity required the following steps:

- Registering for a Splunk account and completing the certification registration

if interested:

- Complete the remaining courses in your own time.
- Take the Splunk Core Certified User exam to earn the certificate.

Note the following resources for taking the exam:

- [Splunk: Core Certified User certification](#)
- [Splunk: Video on getting certified](#)

- [Splunk: Exam registration tutorial](#)
 - [Pearson VUE: Splunk exam registration](#)
 - [Splunk: Certification study guide](#)
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11. Overview of Remaining Projects and Introduction to BootCon

You will complete two more projects in this course:

- **Defensive Security Project:** This will take place at the end of the Defensive Security module. Details will be provided at the beginning of the project week.
- **Final Project (bootCon):** This occurs during the last week of class. We'll provide an overview now so that students can begin brainstorming and preparing for their project.

Final Project: BootCon

On the last day of this boot camp, students will attend a mock cybersecurity conference called bootCon.

- During this conference, they will have an opportunity to showcase the skills they've acquired throughout the boot camp.
- Each student will be required to present a cybersecurity topic of their choice during bootCon.

The BootCon Presentation

Note the following about bootCon presentations:

- Cybersecurity professionals commonly present the following to their peers:
 - Security research they're conducting
 - Newly discovered security vulnerabilities of products, devices, software, or hardware
 - Demonstrations of the hacks that will exploit these vulnerabilities
 - Mitigations to protect against these vulnerabilities
- These presentations often occur at security conferences, trade shows, and other industry events. Cybersecurity professionals commonly attend these events to remain abreast of technological developments and new techniques.
- Similarly, graduates of the Cybersecurity boot camp will have an opportunity to showcase the skills they've learned with a bootCon presentation.

- Most cyber professionals present at conferences when they've found a new vulnerability. However, for bootCon presentations, it is acceptable and recommended that students recreate a finding that has already been discovered.
- A bootCon presentation is NOT a research paper.
- While research will be required, all presentations must be tangible and demonstrable.
- A demonstration can either be conducted in person or in a prerecorded video that accompanies the presentation if a live demonstration isn't practicable.

Each bootCon presentation should fall into one of the following three categories:

1. Exploiting a vulnerability of an IoT device
 - For example, hacking your personal Blu-ray player
2. Developing code or a program that can complete a cybersecurity task
 - For example, developing a Python script that can automate an Nmap scan
3. Demonstrating how a cybersecurity tool that was not covered in class can accomplish a specific goal
 - For example, using SET to design a social engineering campaign.

BootCon Presentation Rules and Requirements

Note the following rules and requirements for bootCon presentations:

1. Students must submit a project summary to their instructor for approval before proceeding. The summary should be submitted with a Slack message by Week 22. (*Instructor note: Provide the date for your class.*)

The project summary should include:

- Topic and title of your presentation
 - End goal or vulnerability being exploited
 - List of devices and/or technologies that will be used to accomplish the goal
 - Summary of how the devices and technologies will be used to accomplish the goal
 - *Instructor note: Refer to the following guide to approve your students' bootCon proposals: [BootCon Presentation Approval Guide](#)*
2.  UNDER NO CIRCUMSTANCE MAY ANY ASPECT OF A STUDENT'S BOOTCON PRESENTATION BE UNETHICAL OR ILLEGAL. 
 - You must perform all hacks and tests in simulated environments.
 - You must complete any network connections in your home and/or controlled environment.

- You may only perform IoT hacks on devices that you own.
- 3. Presentations must have a goal whose achievement you can demonstrate.
 - For example:
 - Goal: Cracking WEP wireless traffic from your home router
 - Demonstration: Demonstrating how you captured and cracked your wireless traffic
 - You can either conduct your demonstration live or record it and present it while you walk through what took place.
- 4. Students must submit their presentation in the form of a Google Slides deck that, at a minimum, includes the following:
 - Cover slide: Presentation title and team member(s) presenting
 - Technical background:
 - Explanation of why you selected the topic you are presenting
 - Networking, cryptographic, or security concepts applied
 - Research steps taken
 - Demonstration preview: Preview of the steps that you'll take in the upcoming demonstration
 - Demonstration: A live or recorded demonstration
 - Demonstration summary: Summary of the demonstration and its impact
 - Mitigation: Recommendations for mitigating against the attack you just conducted (if your presentation isn't about an attack, this is not required)
- 5. Students will present their project on the last day of class, and their total presentation time should be 7–10 minutes.
 - *Instructor note: This is our suggested time guidance, but select the appropriate time per presentation based on your class size.*
 - If you choose to form a group of 2–3 students, presentation time can be 5 minutes per student.

BootCon Presentation Ideas and Resources

For any students who need ideas for their bootCon presentation, we have created a document that contains:

- Sample presentations
- List of Kali Linux tools
- List of IoT hacks
- Videos of hacks presented at security conferences

Note the following document: [bootCon Presentation Guide](#)

- This document contains a set of sample instructions for a bootCon presentation.

- Remind students that it is completely acceptable to recreate an existing presentation, as long as they are able to demonstrate the concept themselves.

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